Computational Validation and Structured-Mesh Analysis of Transonic Flow over the RAE 2822 Airfoil Using the k–omega SST Turbulence Model

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Abstract

Transonic flow over an airfoil produces localized supersonic regions, shock waves, and shock-wave/boundary-layer interactions that influence pressure distribution, lift, drag, and flow separation. This study develops a two-dimensional computational model of the RAE 2822 benchmark airfoil using ANSYS ICEM CFD and ANSYS Fluent. A structured mesh and the k-omega SST turbulence model are used to predict surface pressure coefficient, Mach-number contours, aerodynamic forces, and convergence behavior. Preliminary simulations capture upper-surface acceleration, shock formation, and pressure recovery, but quantitative agreement with benchmark data has not yet been achieved. Current work focuses on improving mesh topology, near-wall spacing, wake refinement, and boundary-condition treatment. After validation, the computational workflow will be applied to shock-control bump geometries to evaluate their effects on shock strength, flow separation, drag, and aerodynamic efficiency.

Background

- During transonic flight, parts of the airflow over a wing may become supersonic even when the aircraft remains below Mach 1.
- When this supersonic flow slows back to subsonic speed, a shock wave forms and causes a rapid pressure increase.
- Shock waves can increase wave drag, interact with the boundary layer, and lead to flow separation.
- Accurate prediction of shock location and pressure recovery is important for improving aerodynamic efficiency and aircraft performance.
- The RAE 2822 airfoil is widely used as a validation benchmark because published experimental pressure and boundary-layer data are available.
- Validating the baseline airfoil model provides confidence before testing shock-control bumps designed to weaken or redistribute the shock.

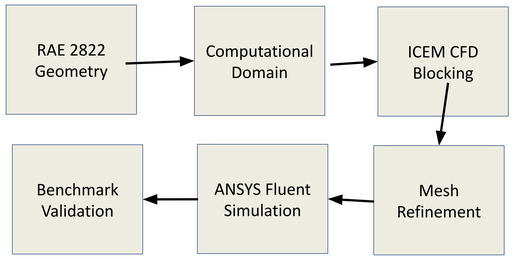
Research Question & Hypothesis

Research question: Can a refined structured mesh generated in ANSYS ICEM CFD, combined with the k-omega SST turbulence model in ANSYS Fluent, accurately reproduce the published pressure distribution and shock location for the RAE 2822 airfoil?

Hypothesis: Improving mesh topology, near-wall spacing, and wake refinement will produce closer agreement with experimental pressure-coefficient data and more stable convergence than the preliminary mesh. After validation, shock-control bumps placed near the baseline shock location are expected to weaken or redistribute the shock and reduce wave drag without substantially decreasing lift.

Methodology & Computational Setup

- Generated a two-dimensional RAE 2822 airfoil geometry and placed it inside a far-field computational domain.
- Created a structured mesh in ANSYS ICEM CFD with refinement near the airfoil surface, leading edge, trailing edge, shock region, and downstream wake.
- Simulated compressible transonic flow in ANSYS Fluent using the k-omega SST turbulence model at Mach 0.75, an angle of attack of 3.19 degrees, and a Reynolds number of 6.2 million.
- Generated surface pressure coefficient, Mach-number contours, lift, drag, residuals, and wall y-plus for future comparison with published benchmark data.



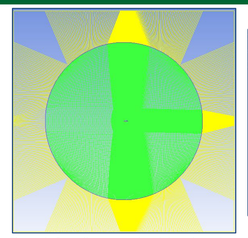


Figure 1. Overall computational domain and structured blocking topology generated in ANSYS ICEM CFD for the RAE 2822 airfoil.

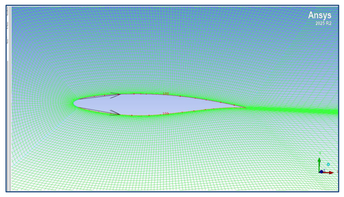


Figure 2. Close-up of the structured mesh showing refinement near the airfoil surface, leading edge, trailing edge, and downstream wake.

Computational Setup	
Parameter	**Simulation Setup**
Airfoil geometry	RAE 2822
Model dimensionality	Two-dimensional
Freestream Mach number	0.75
Angle of attack	3.19°
Reynolds number	6.2 million
Turbulence model	k-omega SST
Meshing software	ANSYS ICEM CFD 2025 R2
Flow solver	ANSYS Fluent 2025 R2
Airfoil boundary	No-slip wall
Primary validation output	Surface pressure coefficient
Additional outputs	Mach contours, lift, drag, residuals
Published reference mesh	545,768 cells

Preliminary Simulation Results

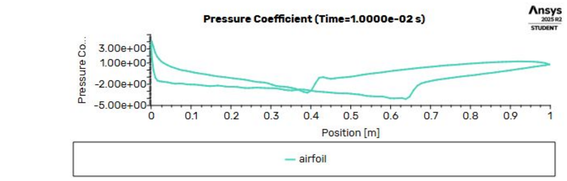


Figure 3. Preliminary surface pressure-coefficient distribution over the RAE 2822 airfoil, plotted against normalized chordwise position from the leading edge, x/c = 0, to the trailing edge, x/c = 1.

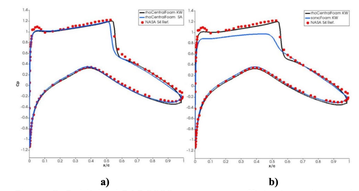


Figure 8. Published RAE 2822 pressure-coefficient results used as the validation target. Adapted from Malkocoglu et al. [1].

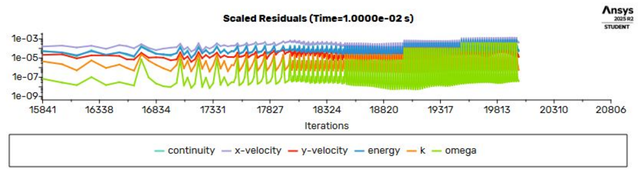


Figure 5. Scaled continuity, velocity, energy, k, and omega residuals plotted against solver iteration number.

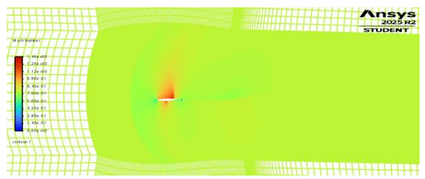


Figure 4. Preliminary Mach-number contour showing localized flow acceleration near the airfoil, shock formation, and the downstream wake.

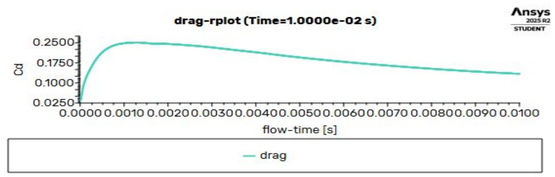


Figure 6. Drag-coefficient history plotted against simulated flow time in seconds.

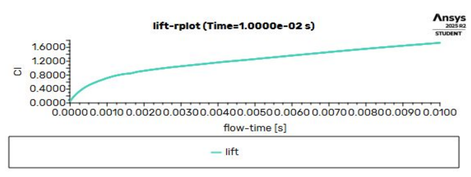


Figure 7. Lift-coefficient history plotted against simulated flow time in seconds.

- The preliminary solution captures upper-surface acceleration, shock formation, and pressure recovery.
- The pressure-coefficient distribution does not yet quantitatively match the benchmark data.
- Oscillations in residuals, lift, and drag indicate that additional convergence and mesh refinement are required.
- The current results guide improvements in near-wall spacing, trailing-edge treatment, wake refinement, and boundary conditions.

Conclusions

- A complete CFD workflow was established for the RAE 2822 airfoil using ANSYS ICEM CFD and ANSYS Fluent.
- Preliminary results capture key transonic features, including upper-surface acceleration, shock formation, pressure recovery, lift, and drag.
- The current solution is not yet quantitatively validated because mesh quality and convergence still affect the pressure distribution and force histories.
- Mesh topology, near-wall spacing, trailing-edge treatment, wake alignment, and boundary conditions strongly influence the final solution.

Future Directions

- Refine the structured mesh to improve near-wall resolution, shock capture, and wake quality.
- Rerun the baseline case at Mach 0.75, 3.19 degrees angle of attack, and Reynolds number 6.2 million.
- Compare pressure coefficient, shock location, lift, drag, residuals, and y-plus with published benchmark data.
- Perform a mesh-sensitivity study to confirm that the results are independent of grid resolution.
- Introduce shock-control bumps and evaluate their effects on shock strength, drag, lift, and aerodynamic efficiency.

References

[1] Malkocoglu, U. E., et al., “Experimental and Computational Investigations of Shock Wave–Boundary Layer Interaction on a Transonic Airfoil with and without Bump,” *AIAA AVIATION 2020 Forum*, AIAA Paper 2020-2759, 2020.

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[6] NASA NPARC Alliance, “RAE 2822 Transonic Airfoil: Study No. 4,” *CFD Verification and Validation Archive*, NASA Glenn Research Center, 2002.

Acknowledgments

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